IRRIGATION WATER MANAGEMENT · NOTE WM03

Hydraulics of Pipe and Open-Channel Flow — Continuity, Bernoulli, Friction, Chezy and Manning

Course topics: IDE3.6 L10, IDE3.6 L11, IDE-509 T12

Pipes and channels

- Flow in channels and pipes is hydraulically similar. Pipe-flow equations deal with differences in pressure head and elevation head; in an open channel the pressure head does not change and the slope of the water surface drives the flow.
- A flowing fluid meets resistance from viscosity and friction. At low velocity it moves in bands parallel to the walls — **viscous (laminar) flow**; at high velocity cross-currents and eddies form, with more resistance.
- Velocity is not uniform across a section (slower near the sides), but engineers assume a uniform velocity equal to the mean.
- Any obstruction, change of section or change of direction disturbs steady flow, creates eddies and adds energy loss.

Basic equations

$$\text{Continuity: } Q = a_1 v_1 = a_2 v_2$$

The volume rate of flow past every successive cross-section is the same at any instant.

$$\text{Bernoulli (total energy): } E = p/w + v^2/2g + z$$

p = unit pressure; w = unit weight of water; v = mean velocity; z = elevation head. Bernoulli's theorem applies the conservation of energy to flowing water: without friction the sum of the three heads is constant. With losses between an upstream section 1 and a downstream section 2:

$$v_1^2/2g + p_1/w + z_1 = v_2^2/2g + p_2/w + z_2 + h_f + h_l$$

h_f = friction loss between the sections; h_l = other losses of pressure head.

Friction loss in pipes

$$h_f = f\, l\, v^2 \,/\, (2\, g\, R);\ \text{for a circular pipe running full } R = d/4,\ \text{so } h_f = 4\, f\, l\, v^2 \,/\, (2\, g\, d)$$

- f = coefficient of friction (it covers eddy resistance as well as wall friction, and is lower for new, smooth pipes); l = length; v = velocity; R = hydraulic radius — units generally metres and seconds.
- Empirical values for steel pipes (d in cm): old pipes f = 0.01 (1 + 1/(4.725 d)); new pipes f = 0.005 (1 + 1/(4.725 d)). Because the 1/(4.725 d) term is very small it is usually dropped: f ≈ 0.01 (old) and 0.005 (new).

Hydraulic and energy gradients

- **Hydraulic grade line:** the water surface in an open channel; in a pipe, the line joining the levels to which water rises in piezometer tubes.
- **Energy grade line:** lies above the hydraulic grade line by the velocity head $v^2/2g$; its fall over a length is the friction loss.
- In a straight uniform pipe the hydraulic gradient is a straight line; otherwise find its points successively with the head-loss equation.

$$h_l = p_1/w - p_2/w = f\,l\,v^2/(2\,g\,R);\ \text{hydraulic gradient} = \text{head lost by friction} / \text{total length of pipe} = h_f / L$$

Pipe flow exists when a closed conduit flows full: the conduit fixes the area, the water is not open to the air, and the pressure may be equal to, above or below atmospheric. It applies to culverts, drop inlets, siphons and all pipelines.

Open channels

An open channel is any conduit with a free water surface — rivers, canals, field ditches, uncovered flumes, and pipes or drains running part full. Water flows from higher to lower ground under gravity.

Element (trapezoidal channel, Fig. 7.13)	Symbol
Top width of channel / width of water surface at depth d	T / t
Depth of channel including freeboard / depth of flow	D / d
Bottom width	b
Wetted sides	a, c
Angle between side and horizontal	θ

- **Wetted perimeter** p = a + b + c — the surface in contact with water. Friction depends on it: for the same area, a larger p gives a lower velocity. Experimentally $v \propto \sqrt{a}$ and $v \propto 1/\sqrt{p}$.
- **Hydraulic radius** R = a/p; $v \propto \sqrt{R}$, and R grows as p shrinks for a given area.
- **Hydraulic slope** S = h/l (vertical drop over length); $v \propto \sqrt{S}$.
- **Freeboard** — vertical distance from the highest expected water level to the top of the banks, against overtopping by waves or the unforeseen.
- **Angle of repose** — the steepest stable slope of a material; steeper sides slip, move in mass or erode.

$$\text{Chezy: } v = C\sqrt{(R\,S)}\ \text{Manning (preferred): } v = R^{(2/3)}\,S^{(1/2)} / n\ \text{Capacity: } Q = a\,v$$

- From the proportionalities, $v \propto \sqrt{(R\,S)}$. Chezy's C depends on channel size, slope, depth and surface, and is hard to fix; Manning's n is the roughness coefficient of the lining (values in Appendix F of the textbook), and the two give closely comparable results.
- v in m/s, R in m, S = h/l, Q in m^3/s, a in m^2.

> A lined channel: bed 0.5 m, depth 0.3 m, side slopes 1 : 1, S = 0.001, n = 0.015. Area = (0.5 + 0.3) × 0.3 = 0.24 m^2; p = 0.5 + 2 × 0.3 × √2 = 1.349 m; R = 0.178 m. v = 0.178^(2/3) × 0.001^(1/2) / 0.015 = 0.316 × 0.0316 / 0.015 ≈ 0.67 m/s; Q ≈ 0.16 m^3/s.

Sources: ICAR eCourse 'Irrigation Engineering' — Dr K. N. Tiwari and Dr N. S. Raghuwanshi (IIT Kharagpur), reviewed by Dr S. D. Gorantiwar (MPKV Rahuri). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Hydraulic radius is

(a) area × wetted perimeter
(b) area / wetted perimeter
(c) wetted perimeter / area
(d) depth / width

Q2. In Manning's formula velocity varies as

(a) R^(1/2) S^(2/3)
(b) R^(2/3) S^(1/2)
(c) R S
(d) R^2 S

Q3. For a circular pipe running full, R equals

(a) d
(b) d/2
(c) d/4
(d) 2d

Q4. The energy grade line lies above the hydraulic grade line by

(a) the friction head
(b) the pressure head
(c) the velocity head
(d) the elevation head

Q5. Bernoulli's theorem is an application of the conservation of

(a) mass
(b) momentum
(c) energy
(d) charge

Answers

Q1 — (b) area / wetted perimeter
Q2 — (b) R^(2/3) S^(1/2)
Q3 — (c) d/4
Q4 — (c) the velocity head
Q5 — (c) energy